



Linear Models V

Megan Ayers
 Stat 141 | Fall 2026
 Friday, Week 5

Goals for Today

- Some logistics - Megan will be away for a few days coming up
- Handling varying slopes
- Transformations
- Regression with many variables
- Adjusted R^2 and model selection
- Multicollinearity
- (Ambitious) Application of Multiple Linear Regression

Office hour change next week

- Megan's Wednesday office hours will be on **Tuesday** next week 10/6
- Double check Moodle!

Logistics for Thurs Oct 8

- Quiz 4 will be taken online on Gradescope at your lab section's start time
- It will be timed like usual, but you can take it anywhere (you still need to complete it individually)
- There will be a short metacognition homework assignment, but no lab assignment
- There won't be a quiz on Oct 15

Friday Oct 9 and Monday Oct 12

- Lenny will lecture on these days
- No office hours with Megan these days

Linear Regression

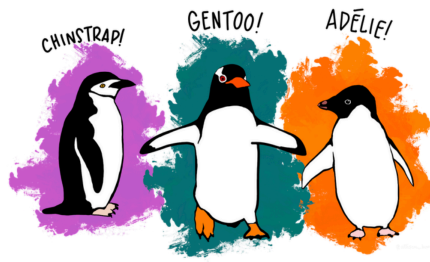
Model Form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

Linear regression is a flexible class of models that allow for:

- Both quantitative and categorical **explanatory** variables.
- **Multiple** explanatory variables.
- Transformed explanatory variables (we'll see this today).
- BUT the **response variable is quantitative**.

More time with the palmerpenguins!



The Palmer Archipelago penguins. Artwork by @allison_horst.

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Contrasting model forms

Recall the equal slopes model:

$$y = \beta_0 + \beta_1 \cdot \text{XBill Depth} + \beta_2 \cdot \text{XSpecies:Chinstrap} + \beta_3 \cdot \text{XSpecies:Gentoo} + \epsilon$$

How can we allow the slopes to vary?

$$y = \beta_0 + \beta_1 \cdot \text{XBill Depth} + \beta_2 \cdot \text{XSpecies:Chinstrap} + \beta_3 \cdot \text{XSpecies:Gentoo} + \beta_4 \cdot \text{XBill Depth} \cdot \text{XSpecies:Chinstrap} + \beta_5 \cdot \text{XBill Depth} \cdot \text{XSpecies:Gentoo} + \epsilon$$

• Coefficient interpretation?

- Those involving XBill Depth are particularly tricky
- Ex. β_4 : When bill depth increases by 1mm for a *Chinstrap* penguin, we expect bill length to change by β_4 *more/less than it would change* for an *Adelie* penguin.

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Different Slopes Model in R

```
1 same_slope <- lm(bill_length_mm ~ bill_depth_mm + species, penguins)
2
3 diff_slope <- lm(bill_length_mm ~ bill_depth_mm * species, penguins)
```

```
1 get_regression_table(same_slope)
# A tibble: 4 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 13.2      2.25     5.88    0      8.80    17.6
2 bill_depth_mm 1.39    0.122    11.4    0      1.15    1.63
3 species: Chinstrap 9.94    0.368    27.0    0      9.22    10.7
4 species: Gentoo 13.4     0.512    26.2    0     12.4    14.4

1 get_regression_table(diff_slope)
# A tibble: 6 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 23.1      3.02     7.65    0     17.1    29.0
2 bill_depth_mm 0.857    0.164     5.22    0     0.534    1.18
3 species: Chinstrap -9.64    5.72    -1.69   0.093   -20.9    1.60
4 species: Gentoo -5.84    4.54    -1.29   0.199   -14.8    3.08
5 bill_depth_mm:speciesC... 1.06    0.31     3.44   0.001    0.455    1.68
6 bill_depth_mm:speciesG... 1.16    0.279     4.17    0     0.615    1.71
```

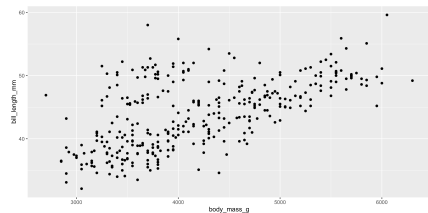
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Regression with multiple quantitative predictors

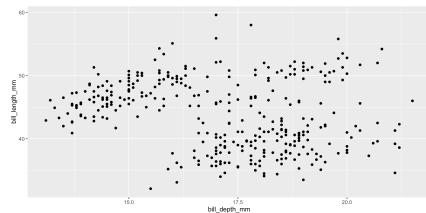
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Regression with multiple quantitative predictors

```
1 ggplot(penguins, aes(x = body_mass_g, y = bill_length_mm))
2 geom_point()
```



```
1 ggplot(penguins, aes(x = bill_depth_mm, y = bill_length_mm))
2 geom_point()
```



$$y = \beta_0 + \beta_1 X_{\text{Bill Depth}} + \beta_2 X_{\text{Body Mass}} + \epsilon$$

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Regression with multiple quantitative predictors

In R:

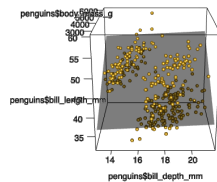
```
1 mod <- lm(bill_length_mm ~ bill_depth_mm + body_mass_g, penguins)
2 get_regression_table(mod)

# A tibble: 3 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 23.3      3.27      7.14    0      16.9    29.7
2 bill_depth_mm 0.163    0.137     1.19   0.234  -0.106  0.432
3 body_mass_g 0.004      0        12.6    0      0.004   0.005
```

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Visualizing the best fit plane

```
1 options(rgl.useNULL = TRUE)
2 options(rgl.printRglwidget = TRUE)
3 library(rgl)
4 plot3d(x = penguins$bill_depth_mm, y = penguins$body_mass_g,
5        z = penguins$bill_length_mm, type = "s", col = "goldenrod", size = 1)
6 planes3d(a = coef(mod)[2], b = coef(mod)[3], c = -1, d = coef(mod)[1], alpha = .5)
```



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Least Squares and Prediction

- Still minimizing the sum of squared residuals (from the plane)
- Still predict like usual

```
1 pred_dat <- data.frame(bill_depth_mm = c(16, 18),
2                        body_mass_g = c(4000, 5000))
3 predict(mod, pred_dat)

1      2
42.87888 47.44527
```

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Least Squares and Interpretation (Generally)

```
1 mod <- lm(bill_length_mm ~ bill_depth_mm + body_mass_g, penguins)
2 get_regression_table(mod)

# A tibble: 3 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept    23.3        3.27        7.14     0      16.9    29.7
2 bill_depth_mm  0.163       0.137        1.19    0.234   -0.106  0.432
3 body_mass_g   0.004        0         12.6     0      0.004  0.005
```

- **Intercept Coefficient:** The *expected/predicted* value of the response, *on average*, when all explanatory variables are *set to 0*.
- **Slope Coefficient on Quantitative Variable:** The *expected/predicted* change in response variable, *on average*, per 1 unit increase in the explanatory variable, *holding all other variables constant*.

If we also had a categorical variable in our model...

- Interpretation on factors from **Categorical Variables:** the *expect/predicted* change in response variable, *on average*, when going from the reference level to the factor's level, *holding all other variables constant*.

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Detour: handling curved relationships

New data context: movie ratings

```
1 library(tidyverse)
2 movies <- read_csv("https://www.lock5stat.com/datasets2e/HollywoodMovies.csv")
3
4 # Restrict our attention to dramas, horrors, and actions
5 movies2 <- movies %>%
6   filter(Genre %in% c("Drama", "Horror", "Action")) %>%
7   drop_na(Genre, AudienceScore, RottenTomatoes)
8 glimpse(movies2)

Rows: 313
Columns: 16
$ Movie      <chr> "Spider-Man 3", "Transformers", "Pirates of the Carib...
$ LeadStudio <chr> "Sony", "Paramount", "Disney", "Warner Bros", "Warner...
$ RottenTomatoes <dbl> 61, 57, 45, 60, 20, 79, 35, 28, 41, 71, 95, 42, 18, 2...
$ AudienceScore <dbl> 54, 89, 74, 90, 68, 86, 55, 56, 81, 52, 84, 55, 70, 6...
$ Story      <chr> "Metamorphosis", "Monster Force", "Rescue", "Sacrific...
$ Genre      <chr> "Action", "Action", "Action", "Action", "Action", "Ac...
$ TheatersOpenWeek <dbl> 4252, 4011, 4362, 3103, 3778, 3408, 3959, 3619, 2911,...
$ OpeningWeekend <dbl> 151.1, 70.5, 114.7, 70.9, 49.1, 33.4, 58.0, 45.3, 19...
$ BOAvgOpenWeekend <dbl> 35540, 17577, 26302, 22844, 12996, 9791, 14663, 12541...
$ DomesticGross <dbl> 336.53, 319.25, 309.42, 210.61, 140.13, 134.53, 131.9...
$ ForeignGross  <dbl> 554.34, 390.46, 654.00, 245.45, 117.90, 249.00, 157.1...
$ WorldGross    <dbl> 890.87, 709.71, 963.42, 456.07, 258.02, 383.53, 289.0...
```

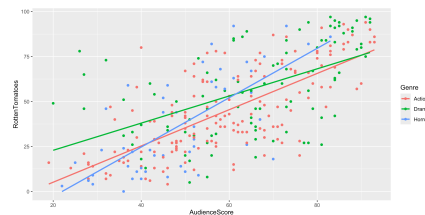
Response variable: RottenTomatoes

Explanatory variables: AudienceScore

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Exploring the Data

```
1 ggplot(data = movies2,
2       mapping = aes(x = AudienceScore,
3                     y = RottenTomatoes,
4                     color = Genre)) +
5   geom_point() +
6   geom_smooth(method = "lm", se = FALSE)
```

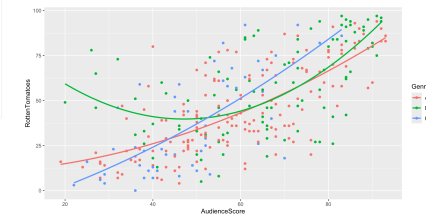


- **Q:** What do the trends suggest, would it make sense to include varying slopes in the model?
- **Q:** Does anyone spot a curved relationship for one of the genres?

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Adding a Curve to your Scatterplot

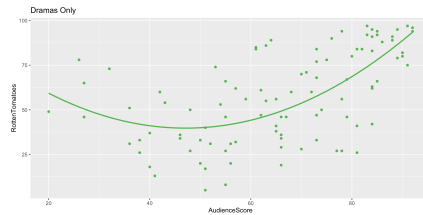
```
1 ggplot(data = movies2,
2       mapping = aes(x = AudienceScore,
3                     y = RottenTomatoes,
4                     color = Genre)) +
5   geom_point() +
6   geom_smooth(method = "lm", se = FALSE,
7               formula = y ~ poly(x, degree = 2))
```



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Adding a Curve to your Scatterplot

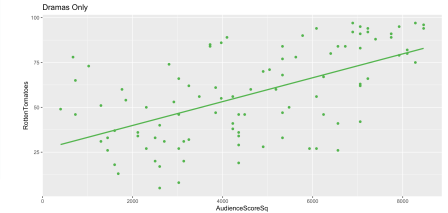
```
1 # Filtering to only Dramas
2 dramas <- movies2 %>% filter(Genre == "Drama")
3
4 ggplot(data = dramas,
5       mapping = aes(x = AudienceScore,
6                     y = RottenTomatoes)) +
7   geom_point(color = "#55B84F") +
8   geom_smooth(method = "lm", se = FALSE,
9             color = "#55B84F",
10            formula = y ~ poly(x, degree = 2)) +
11   ggtitle("Dramas Only")
```



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Using Transformations to Account for Non-Linear Relationships

```
1 # Adding a column for AudienceScore^2
2 dramas <- dramas %>%
3   mutate(AudienceScoreSq = AudienceScore^2)
4
5 ggplot(data = dramas,
6       mapping = aes(x = AudienceScoreSq,
7                     y = RottenTomatoes)) +
8   geom_point(color = "#55B84F") +
9   geom_smooth(method = "lm", se = FALSE,
10             color = "#55B84F") +
11   ggtitle("Dramas Only")
```



- Notice that the relationship between AudienceScore^2 and RottenTomatoes is approximately linear.
- AudienceScore^2 is a **transformation** of AudienceScore

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Fitting a Model with a Transformed Variable

```
1 dramas <- movies2 %>% filter(Genre == "Drama")
2 mod2 <- lm(RottenTomatoes ~ poly(AudienceScore, degree = 2, raw = TRUE),
3           data = dramas)
4 get_regression_table(mod2)
```

A tibble: 3 × 7

term	estimate	std_error	statistic	p_value	lower_ci	upper_ci
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 intercept	98.8	21.3	4.64	0	56.6	141.
2 poly(AudienceScore, de_	-2.51	0.722	-3.48	0.001	-3.94	-1.08
3 poly(AudienceScore, de_	0.027	0.006	4.58	0	0.015	0.038

- We'll practice a very common transformation in HW 5 - the **log transformation**
- **Q:** Why is it still called a *linear* regression if the model also handles curved relationships?
- Transformations are super powerful: we can now use linear models even if the *untransformed* variables have non-linear relationships.

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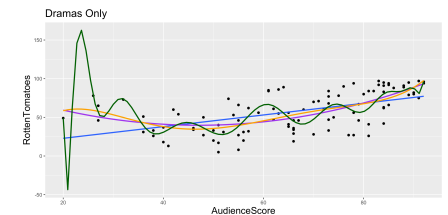
Model Building Guidance

What degree of polynomial should I include in my model?

Guiding Principle: Capture the general trend, not the noise.

$$y = f(x) + \epsilon$$

$$y = \text{TREND} + \text{NOISE}$$



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Model Building Guidance

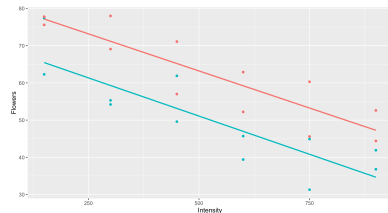
Should we always include an interaction term? How many variables should I include?

Guiding Principle: Occam's Razor for Modeling

"All other things being equal, simpler models are to be preferred over complex ones." – ModernDive

Guiding Principle: Consider your modeling goals.

- The equal slopes model allows us to control for the intensity of the light and then see the impact of being in the early or late timing groups on the number of flowers.
- Later in the course will learn statistical procedures for determining whether or not a particular term should be included in the model.



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What if I want to include many explanatory variables??

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Model Building Guidance

We often have several potential explanatory variables. How do we determine which to include in the model and in what form?

Guiding Principle: Include explanatory variables that attempt to explain **different** aspects of the variation in the response variable.

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Example: Movie Ratings

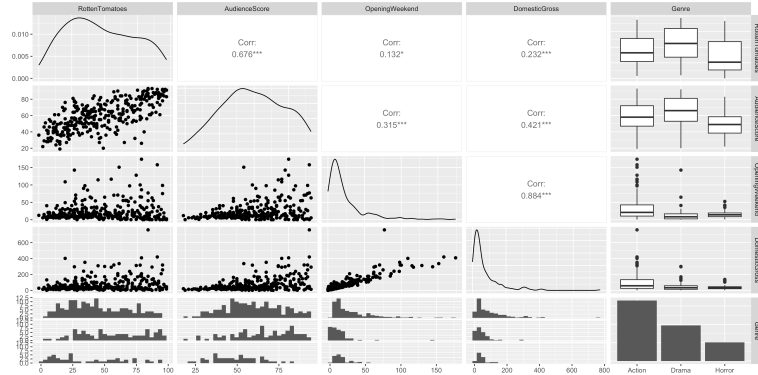
```
1 library(tidyverse)
2 library(moderndive)
3 movies <- read_csv("https://www.lock5stat.com/datasets2e/HollywoodMovies.csv")
4
5 # Restrict our attention to dramas, horrors, and actions
6 movies2 <- movies %>%
7   filter(Genre %in% c("Drama", "Horror", "Action")) %>%
8   drop_na(Genre, AudienceScore, RottenTomatoes)
9 glimpse(movies2)
```

```
Rows: 313
Columns: 16
$ Movie      <chr> "Spider-Man 3", "Transformers", "Pirates of the Carib...
$ LeadStudio <chr> "Sony", "Paramount", "Disney", "Warner Bros", "Warner...
$ RottenTomatoes <dbl> 61, 57, 45, 60, 20, 79, 35, 20, 41, 71, 95, 42, 18, 2...
$ AudienceScore <dbl> 54, 89, 74, 90, 68, 86, 55, 56, 81, 52, 84, 55, 70, 6...
$ Story      <chr> "Metamorphosis", "Monster Force", "Rescue", "Sacrific...
$ Genre      <chr> "Action", "Action", "Action", "Action", "Action", "Ac...
$ TheatersOpenWeek <dbl> 4252, 4011, 4362, 3103, 3778, 3408, 3959, 3619, 2911,...
$ OpeningWeekend <dbl> 151.1, 70.5, 114.7, 70.9, 49.1, 33.4, 58.0, 45.3, 19...
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$ DomesticGross <dbl> 336.53, 319.25, 309.42, 210.61, 140.13, 134.53, 131.9...
$ ForeignGross  <dbl> 554.34, 390.46, 654.00, 245.45, 117.90, 249.00, 157.1...
$ WorldGross    <dbl> 890.87, 709.71, 963.42, 456.07, 258.02, 383.53, 289.0...
```

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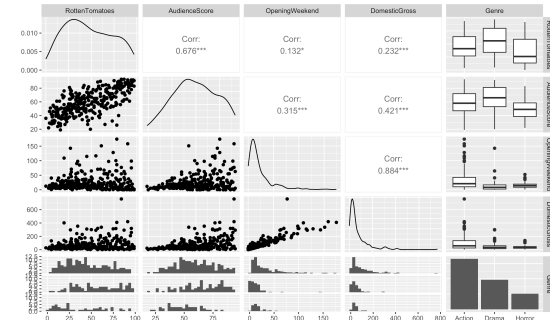
Example: Movie Ratings

```
1 library(GGally)
2 movies2 %>%
3   select(RottenTomatoes, AudienceScore, OpeningWeekend, DomesticGross, Genre) %>%
4   ggpairs()
```



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Example: Movie Ratings



- **Multicollinearity**: when explanatory variables are **highly** correlated with one another. Multicollinearity often results in coefficients that are distorted in erroneous ways!
- **Q**: Do any of these variables seem to be measuring the same thing?

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Model Building Guidance

We often have several potential explanatory variables. How do we determine which to include in the model and in what form?

Guiding Principle: Include explanatory variables that attempt to explain **different** aspects of the variation in the response variable.

```
1 mod_movies <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)
2 get_regression_table(mod_movies)
```

```
# A tibble: 5 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept -12.5        4.08       -3.06    0.002   -20.5    -4.44
2 AudienceScore 0.975      0.072      13.6     0      0.834    1.12
3 Genre: Drama  6.12       2.64       2.31    0.021    0.916   11.3
4 Genre: Horror  2.06       3.14       0.655   0.513   -4.12    8.24
5 DomesticGross -0.006     0.015     -0.431   0.667   -0.035    0.023
```

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Model Building Guidance

Suppose I built 3 different models. **Which is best?**

```
1 mod1 <- lm(RottenTomatoes ~ AudienceScore, data = movies2)
2 mod2 <- lm(RottenTomatoes ~ AudienceScore + Genre, data = movies2)
3 mod3 <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)
```

- Big question! Take **Math 243: Statistical Learning** to learn systematic model selection techniques.
- We will explore one approach. (But there are many possible approaches!)

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Comparing Models with R^2

Strategy: Compute the R^2 value for each model and pick the one with the highest R^2 .

```
1 mod1 <- lm(RottenTomatoes ~ AudienceScore, data = movies2)
2 mod2 <- lm(RottenTomatoes ~ AudienceScore + Genre, data = movies2)
3 mod3 <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)
4
5 get_regression_summaries(mod1) %>% select(r_squared)
# A tibble: 1 × 1
  r_squared
  <dbl>
1      0.457

1 get_regression_summaries(mod2) %>% select(r_squared)
# A tibble: 1 × 1
  r_squared
  <dbl>
1      0.469

1 get_regression_summaries(mod3) %>% select(r_squared)
# A tibble: 1 × 1
  r_squared
  <dbl>
1      0.469
```

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Strategy: Compute the R^2 value for each model and pick the one with the highest R^2 .

```
1 get_regression_summaries(mod1) %>% select(r_squared)
# A tibble: 1 × 1
  r_squared
  <dbl>
1      0.457

1 get_regression_summaries(mod2) %>% select(r_squared)
# A tibble: 1 × 1
  r_squared
  <dbl>
1      0.469

1 get_regression_summaries(mod3) %>% select(r_squared)
# A tibble: 1 × 1
  r_squared
  <dbl>
1      0.469
```

Problem: As we add predictors, the R^2 value will only increase.

Guiding Principle: Occam's Razor for Modeling

"All other things being equal, simpler models are to be preferred over complex ones." –
ModernDive

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Comparing Models with the Adjusted R^2

New Measure of quality: Adjusted R^2 (Coefficient of Determination)

$$\text{adj}R^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2} \left(\frac{n-1}{n-p-1} \right)$$

where p is the number of explanatory variables in the model.

- Now we will penalize larger models.
- **Strategy:** Compute the adjusted R^2 value for each model and pick the one with the highest adjusted R^2 .

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Strategy: Compute the **adjusted R^2** value for each model and pick the one with the highest adjusted R^2 .

```
1 mod1 <- lm(RottenTomatoes ~ AudienceScore, data = movies2)
2 mod2 <- lm(RottenTomatoes ~ AudienceScore + Genre, data = movies2)
3 mod3 <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)
4
5 get_regression_summaries(mod1) %>% select(r_squared, adj_r_squared)
# A tibble: 1 × 2
  r_squared adj_r_squared
  <dbl>      <dbl>
1      0.457      0.455

1 get_regression_summaries(mod2) %>% select(r_squared, adj_r_squared)
# A tibble: 1 × 2
  r_squared adj_r_squared
  <dbl>      <dbl>
1      0.469      0.464

1 get_regression_summaries(mod3) %>% select(r_squared, adj_r_squared)
# A tibble: 1 × 2
  r_squared adj_r_squared
  <dbl>      <dbl>
1      0.469      0.462
```

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Recap on Developing Linear Models

- Key components that we've learned about over the last 5 lectures:
 - Determining the **response** variable and the potential **explanatory** variable(s)
 - Writing out the **model form** and understanding what the coefficients represent **in context**
 - **Building** and **visualizing** linear regression models in **R**
 - **Validating** model assumptions with diagnostic plots
 - **Comparing** different potential models

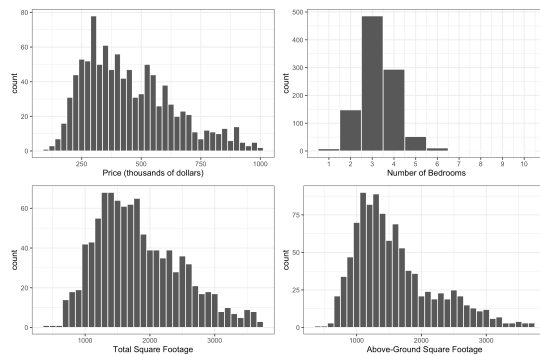
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Case Study

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Case Study: Home Prices in King County, WA

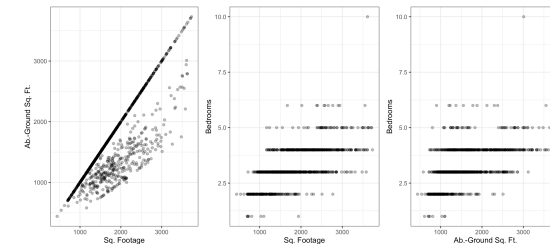
We'll consider : Sale price (**response**); and square footage, above-ground square footage, and number of bedrooms (potential explanatory variables). Check out the exploratory plots below:



- Q: How would you describe the distribution of **price**?
- Q: What do you expect the relationship between **price** and **bedrooms** to be like?

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Exploration, Multicollinearity



- **Multicollinearity**: when explanatory variables are **highly** correlated with one another. Multicollinearity often results in coefficients that are distorted in erroneous ways! We want to avoid this (*some* correlation is okay!)
- Q: Which pair of explanatory variables suffers *most* from multicollinearity?

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Model Estimation

```
# A tibble: 3 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 197.        19.4        10.1     0       159.    235.
2 sqft_living 0.177       0.01        18.4     0       0.158    0.196
3 bedrooms -21.6        7.27        -2.97    0.003   -35.9    -7.34
```

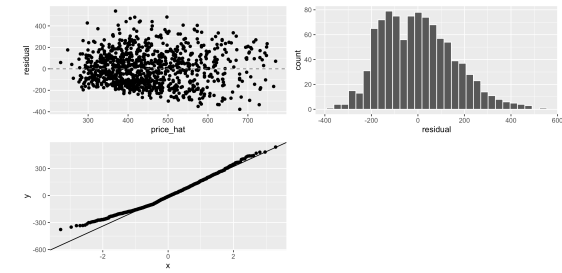
- Q: What is the equation, including coefficient values, for the estimated model?

$$\widehat{\text{price}} = 197 + 0.177 * \text{sqft_living} - 21.6 * \text{bedrooms}$$

- Q: How should interpret the **intercept** coefficient in our model?
 - When **sqft_living** and **bedrooms** are both 0, we expect/predict price to be \$197,000 on average.
- Q: The coefficient for **Bedrooms**? Does this match your expectation?
 - When the number of **bedrooms** increases by 1, we expect/predict price to **decrease** by \$21,600 on average, **holding sqft_living constant**.

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LINE Assumptions



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Adjusted R^2

What's the adjusted R^2 of our model?

```
1 get_regression_summaries(mod)[, c("r_squared", "adj_r_squared")]
# A tibble: 1 × 2
  r_squared adj_r_squared
<dbl>      <dbl>
1 0.314      0.313
```

Not super close to 1. Compare to a simpler model:

```
1 small_mod <- lm(price ~ bedrooms, data = house)
2 get_regression_summaries(small_mod)[, c("r_squared", "adj_r_squared")]
# A tibble: 1 × 2
  r_squared adj_r_squared
<dbl>      <dbl>
1 0.082      0.081
```

Compare to a much bigger model with additional variables:

```
1 big_mod <- lm(price ~ bedrooms + bathrooms + sqft_living + sqft_above + sqft_lot +
2 view + condition + yr_built, data = house)
3 get_regression_summaries(big_mod)[, c("r_squared", "adj_r_squared")]
# A tibble: 1 × 2
  r_squared adj_r_squared
<dbl>      <dbl>
1 0.434      0.429
```

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